

1

What is Microsoft Azure?



ANSWER

Azure is Microsoft's cloud platform. It provides compute, storage, networking, databases, analytics, AI, security and management services from Microsoft datacentres.



2

What is cloud computing?



ANSWER

Cloud computing delivers IT resources over a network on demand. Customers can scale quickly and usually pay for what they consume instead of owning all hardware.



3

What is the difference between IaaS, PaaS and SaaS?



ANSWER

IaaS gives virtual infrastructure, PaaS gives a managed application platform, and SaaS gives a complete application. Customer management decreases from IaaS to SaaS.



4

What are an Azure region and an Availability Zone?



ANSWER

A region is a geographic Azure area. An Availability Zone is a separated datacentre group within a region, with independent power, cooling and networking.



5

What is an Azure resource group?



ANSWER

It is a logical container for related Azure resources. Resources with a common lifecycle are often grouped so access, policy, deployment and deletion can be managed together.



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What is an Azure subscription?

ANSWER

A subscription is a billing and management boundary linked to an Azure account or agreement. Resource quotas, RBAC and policies can be applied at subscription scope.

7

What is a management group?

ANSWER

Management groups organise multiple subscriptions in a hierarchy. Governance settings such as Azure Policy and RBAC can inherit from a management group to its descendants.

8

What is Azure Resource Manager (ARM)?

ANSWER

ARM is Azure's deployment and management layer. It accepts requests from the portal, CLI, PowerShell, SDKs or templates and applies access control, policy, locks and tags.

9

What is Bicep, and how is it related to ARM templates?

ANSWER

Bicep is Azure's declarative infrastructure language. It offers simpler syntax and compiles to ARM template JSON, while using the same ARM resource providers.

10

How can you manage Azure resources?

ANSWER

Common choices are the Azure portal, Azure CLI, Azure PowerShell, REST APIs, SDKs and infrastructure-as-code tools such as Bicep, ARM templates and Terraform.



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11 What is Azure RBAC?

ANSWER

Azure role-based access control authorises actions on Azure resources. A role assignment combines a security principal, a role definition and a scope.

12 What is Microsoft Entra ID?

ANSWER

Microsoft Entra ID, formerly Azure Active Directory, is Azure's cloud identity and access service for users, groups, applications, authentication and conditional access.

13 What is a managed identity?

ANSWER

It is an identity for an Azure resource that is managed by Azure. Applications use it to obtain tokens without storing usernames, passwords or client secrets in code.

14 What is a service principal?

ANSWER

A service principal is an application identity in a tenant. Automation and applications use it for controlled access; authentication can use a certificate, secret or federation.

15 How do Azure RBAC and Azure Policy differ?

ANSWER

RBAC controls who can perform actions. Azure Policy evaluates or enforces what resource configurations are allowed, required, audited or remediated.

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16 What is Azure Key Vault?

ANSWER

Key Vault securely stores and controls access to secrets, encryption keys and certificates. Applications should access it with identities and least-privilege permissions.

17 What is an Azure Virtual Network?

ANSWER

A VNet is a private IP network in Azure. It contains subnets and supports routing, filtering, peering and connectivity to the internet, other VNets and on-premises networks.

18 What is a subnet?

ANSWER

A subnet is an IP address range inside a VNet. It segments workloads so routing, network security and service delegation can be applied at a smaller scope.

19 What is a Network Security Group?

ANSWER

An NSG is a stateful packet filter containing prioritised inbound and outbound rules. It can be associated with a subnet or network interface.

20 What is a user-defined route?

ANSWER

A user-defined route customises traffic paths in an Azure route table. It is commonly used to send traffic through a firewall or network virtual appliance.



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21 What is VNet peering?

ANSWER

VNet peering privately connects two virtual networks over Microsoft's backbone. Peered networks communicate with low latency, but address spaces must not overlap.

22 VPN Gateway or ExpressRoute: what is the difference?

ANSWER

VPN Gateway creates encrypted tunnels over the public internet. ExpressRoute provides private connectivity through a provider and is suited to predictable enterprise links.

23 Azure Load Balancer vs Application Gateway?

ANSWER

Load Balancer distributes Layer 4 TCP or UDP traffic. Application Gateway is a regional Layer 7 web traffic load balancer with HTTP routing, TLS termination and optional WAF.

24 What is Azure Front Door?

ANSWER

Front Door is a global Layer 7 entry service for web applications. It offers anycast routing, acceleration, health-based failover, TLS termination, caching and optional WAF.

25 Private Endpoint vs Service Endpoint?

ANSWER

A Private Endpoint gives a service a private IP in your VNet. A Service Endpoint keeps the service public endpoint but extends VNet identity to it over the Azure backbone.

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26 What is Azure Firewall?



ANSWER

Azure Firewall is a managed, stateful network security service. It centrally filters network and application traffic and can provide threat intelligence and logging.



27 How do you improve virtual-machine availability?



ANSWER

Use Availability Zones when supported, or availability sets for fault and update-domain separation. Add load balancing, health probes, backups and tested recovery plans.



28 What is a Virtual Machine Scale Set?



ANSWER

A VM Scale Set creates and manages a group of load-balanced VMs. It supports consistent images, automatic scaling, health-based repair and rolling upgrades.



29 What are Azure Managed Disks?



ANSWER

Managed Disks are block storage volumes managed by Azure for VMs. Azure handles storage accounts, durability and integration with snapshots, encryption and availability features.



30 VM image vs snapshot: what is the difference?



ANSWER

An image is a reusable source for creating VMs and can include OS and data disks. A snapshot is a point-in-time, read-only copy of a single managed disk.



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31 What is Azure Bastion?

ANSWER

Azure Bastion provides managed RDP and SSH access to VMs through the Azure portal or supported clients, without requiring public IP addresses on those VMs.

32 Which core data services are in an Azure Storage account?

ANSWER

Azure Storage includes Blob storage for objects, Files for managed file shares, Queues for messages and Tables for schemaless key-value data.

33 What are Blob Storage access tiers?

ANSWER

Hot is for frequent access, Cool and Cold reduce storage cost for less-frequent access, and Archive has the lowest storage cost but requires rehydration before reading.

34 Explain LRS, ZRS, GRS and GZRS.

ANSWER

LRS replicates locally, ZRS across zones, GRS to a secondary region, and GZRS combines zone replication in the primary region with geo-replication to a secondary region.

35 Azure Files vs Blob Storage?

ANSWER

Azure Files provides managed SMB or NFS file shares. Blob Storage is object storage accessed through HTTP APIs and is ideal for unstructured data at scale.

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36 What is a Shared Access Signature?

ANSWER

A SAS grants delegated, time-limited access to specific storage resources and operations. Prefer a user-delegation SAS where possible and keep scope and lifetime minimal.

37 What is Azure App Service?

ANSWER

App Service is a managed PaaS for hosting web apps and APIs. It handles infrastructure, deployment slots, scaling, TLS, authentication options and platform maintenance.

38 What is Azure Functions?

ANSWER

Azure Functions is a serverless compute service that runs event-driven code. It supports triggers and bindings and can scale according to the selected hosting plan.

39 ACI vs Azure Container Apps?

ANSWER

Azure Container Instances runs containers without managing VMs. Container Apps adds an application platform with revisions, ingress, autoscaling and microservice features.

40 What is Azure Kubernetes Service?

ANSWER

AKS is Azure's managed Kubernetes service. Azure manages the control plane while customers configure workloads, node pools, networking, identities, security and upgrades.

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41

How does AKS autoscaling work?



ANSWER

The Horizontal Pod Autoscaler adjusts pod replicas from metrics. The cluster autoscaler changes node count when pods cannot be scheduled or nodes stay underused.



42

Azure SQL Database vs SQL Managed Instance?



ANSWER

SQL Database is a highly managed database service for modern cloud apps. Managed Instance offers broader SQL Server instance compatibility for migrations needing instance features.



43

What is Azure Cosmos DB?



ANSWER

Cosmos DB is a globally distributed NoSQL database service with multiple APIs, partitioning, tunable consistency choices and provisioned or serverless throughput models.



44

What is Azure Monitor?



ANSWER

Azure Monitor collects and analyses metrics, logs, traces and alerts from applications and infrastructure. It supports dashboards, workbooks, queries and automated responses.



45

Log Analytics vs Application Insights?



ANSWER

Log Analytics queries data in an Azure Monitor workspace. Application Insights adds application performance monitoring such as requests, dependencies, exceptions and traces.



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Azure Backup vs Azure Site Recovery?

ANSWER

Backup protects recoverable copies of data. Site Recovery replicates supported workloads and orchestrates failover and failback for disaster recovery.

47

What is Azure Advisor?

ANSWER

Advisor analyses Azure configurations and provides recommendations for reliability, security, performance, operational excellence and cost optimisation.

48

How do you control Azure costs?

ANSWER

Use tags, budgets, alerts, Cost Analysis, rightsizing, reservations or savings plans where suitable, autoscaling, schedules, storage lifecycle rules and continuous review.

49

What is an Azure DevOps pipeline?

ANSWER

It is an automated CI/CD workflow defined through YAML or the classic editor. It builds, tests and deploys code by running stages, jobs and tasks on agents.

50

How would you troubleshoot a slow Azure web application?

ANSWER

Check scope and timing, then review Application Insights, dependency latency, exceptions, resource metrics, logs, network path and recent changes. Form a hypothesis, test it and verify recovery.



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